

COLE RICHARDS

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Machine Learning / Data Scientist - B.S. Statistics & Data Science, UCLA (Jun 2026). Applied experience in statistical modeling, signal processing on wearable sensor data, geospatial ML on noisy real-world datasets, and rapid prototyping in Python. I'm a curious and driven problem solver who's comfortable owning algorithms from exploration through validation.

EDUCATION

University of California, Los Angeles — B.S. Statistics & Data Science Sep 2024 – Jun 2026

Selected coursework: Computation & Optimization (102B), Applied Geostatistics (C173), Regression & Data Mining (101C), Statistical Consulting (141XP), Probability (100A)

Santa Barbara City College — A.S. Mathematics

Aug 2022 – May 2024

GPA 3.98 · President's Honor Roll

TECHNICAL SKILLS

ML / AI: scikit-learn, statsmodels, gradient boosting, logistic regression, mixed-effects models, cross-validation, hyperparameter tuning, model evaluation (AUC, precision/recall, calibration), LLM-assisted workflows

Signal processing & time-series: time-series decomposition, autocorrelation, feature extraction from sensor data, multi-modal sensor fusion, longitudinal analysis

Languages & tools: Python (pandas, NumPy, matplotlib, scikit-learn, statsmodels), R (tidyverse, ggplot2, geoR, gstat), SQL, Git, Jupyter

Geospatial: ArcGIS, QGIS, GeoPandas, kriging, spatial validation

Experimentation & stats: experimental design, A/B testing, power analysis, Monte Carlo simulation, statistical validation

Visualization & data systems: matplotlib, Plotly, Streamlit, Supabase, Excel

PROJECTS

Sleep Forecasting from Wearable Signals · Python, pandas, scikit-learn, statsmodels, matplotlib

- Built end-to-end ML pipeline on multi-modal Apple Watch sensor data (HR, HRV, activity, steps, prior sleep) to predict next-night sleep quality (duration, deep/REM minutes, awakenings) — from ingestion through feature engineering, modeling, and validation.
- Engineered time-domain features from continuous physiological signals; applied autocorrelation analysis and lagged-feature construction to capture multi-day carryover effects on sleep.
- Designed evaluation methodology with held-out weeks and time-series cross-validation; compared linear models against gradient boosting, identifying step count and afternoon HR as strongest predictors of REM minutes.

California Wildfire Risk Model · Python, GeoPandas, Plotly, Streamlit

- Built ZIP-level risk classifier on incomplete, noisy spatial data — combining CAL FIRE hazard zones, 30-year fire perimeter history, and vegetation indices across LA County.
- Validated against external ground truth (CA Dept. of Insurance non-renewal rates), analogous to validating sensor algorithms against limited reference data.
- Deploying as interactive Streamlit dashboard for non-technical users.

EXPERIENCE

Co-Founder & COO — Trident Ember Defense

Aug 2025 – Present

- Developed a geospatial risk-scoring model (QGIS, ArcGIS, GeoPandas) integrating WUI zones, Fire Hazard Severity Zones, and vegetation indices; deployed scores to drive ZIP-level customer targeting, contributing to 3 closed installations and \$20K+ in early revenue.
- Designed and implemented an internal data stack on Supabase (CRM, quoting, mapping), centralizing operations data and enabling KPI tracking (leads by risk segment, conversion rate).
- Led field execution for a 3,400 sq ft residential project with an 8-person crew.

Inventory Specialist — UCLA Housing IT

Aug 2024 – Dec 2025

- Designed a structured logging system to replace a static spreadsheet, enabling change history, audit trails, and longitudinal trend analysis across thousands of IT assets.
- Built and maintained reports across 12+ product categories, improving data accuracy for device setup, intake, and lifecycle decisions.

INTERESTS

Surfing · Volleyball · Yoga & meditation · Spearfishing